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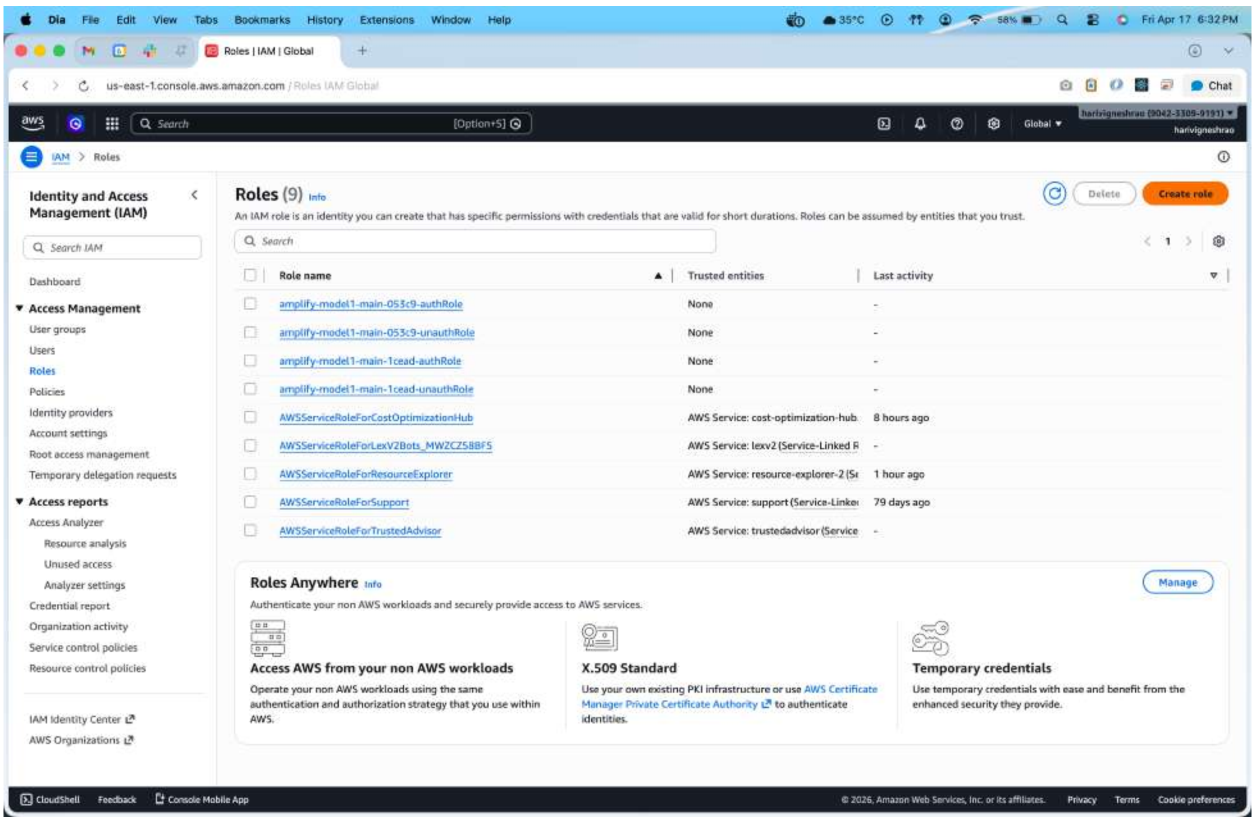
Procedure: Attach an IAM Role to an EC2 Instance (S3 access without keys)

Step 1 — Create Role in Amazon Web Services IAM

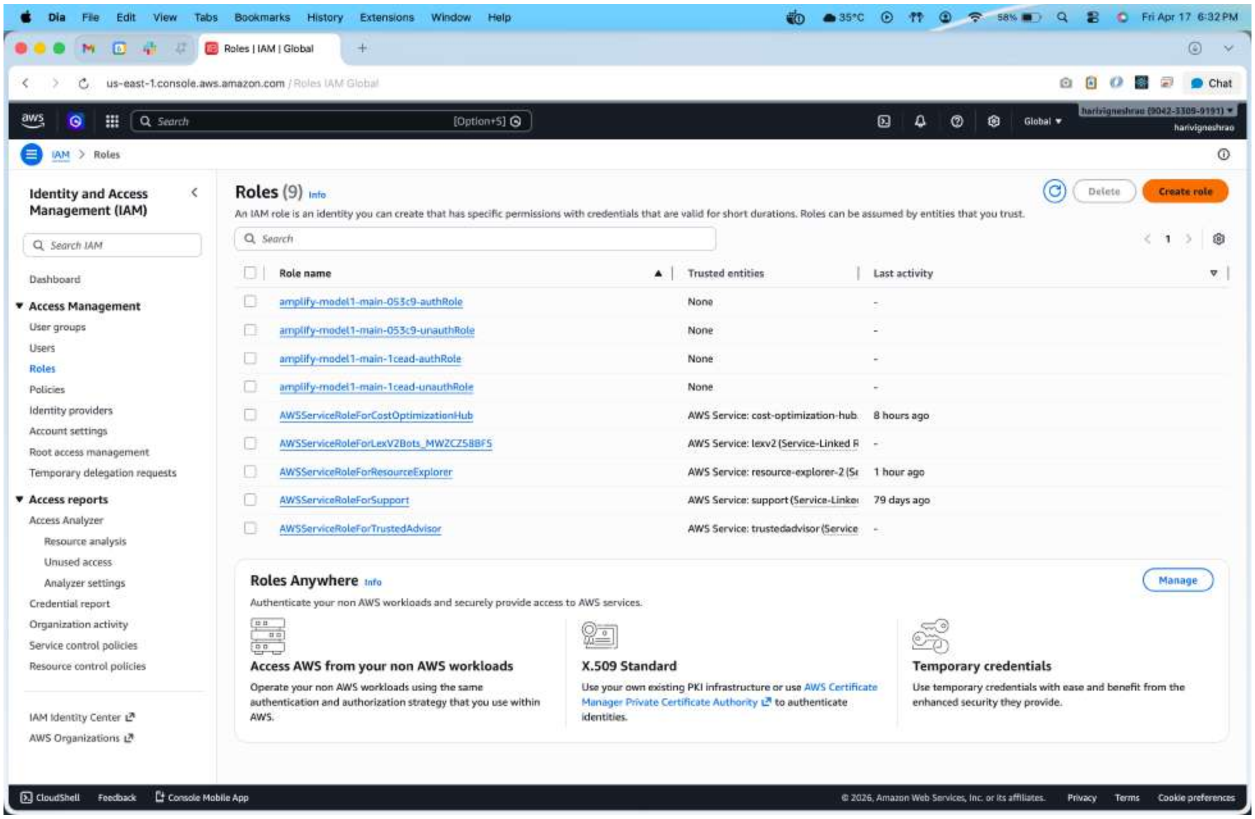
1. Open IAM Console → Roles → Create role



2. Trusted entity: Select AWS Service

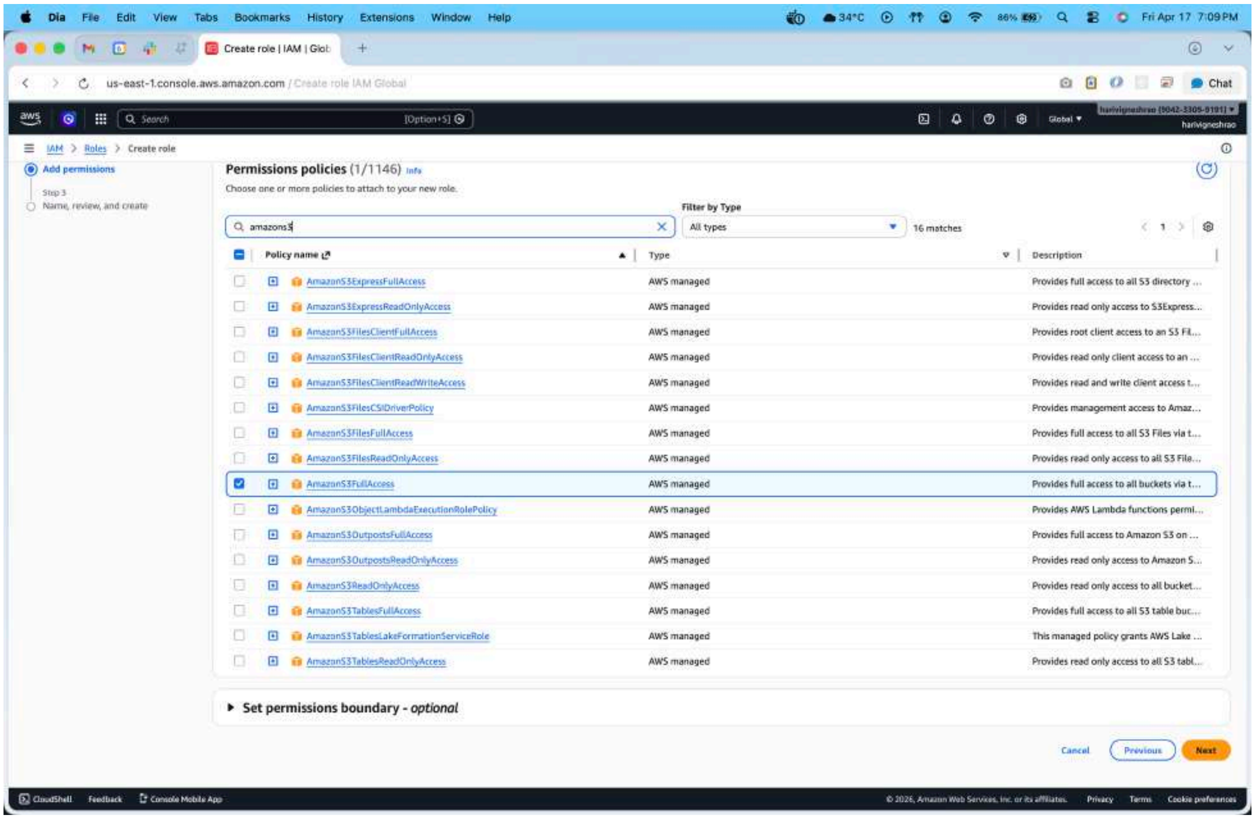
3. Use case: Choose EC2

4. Click Next



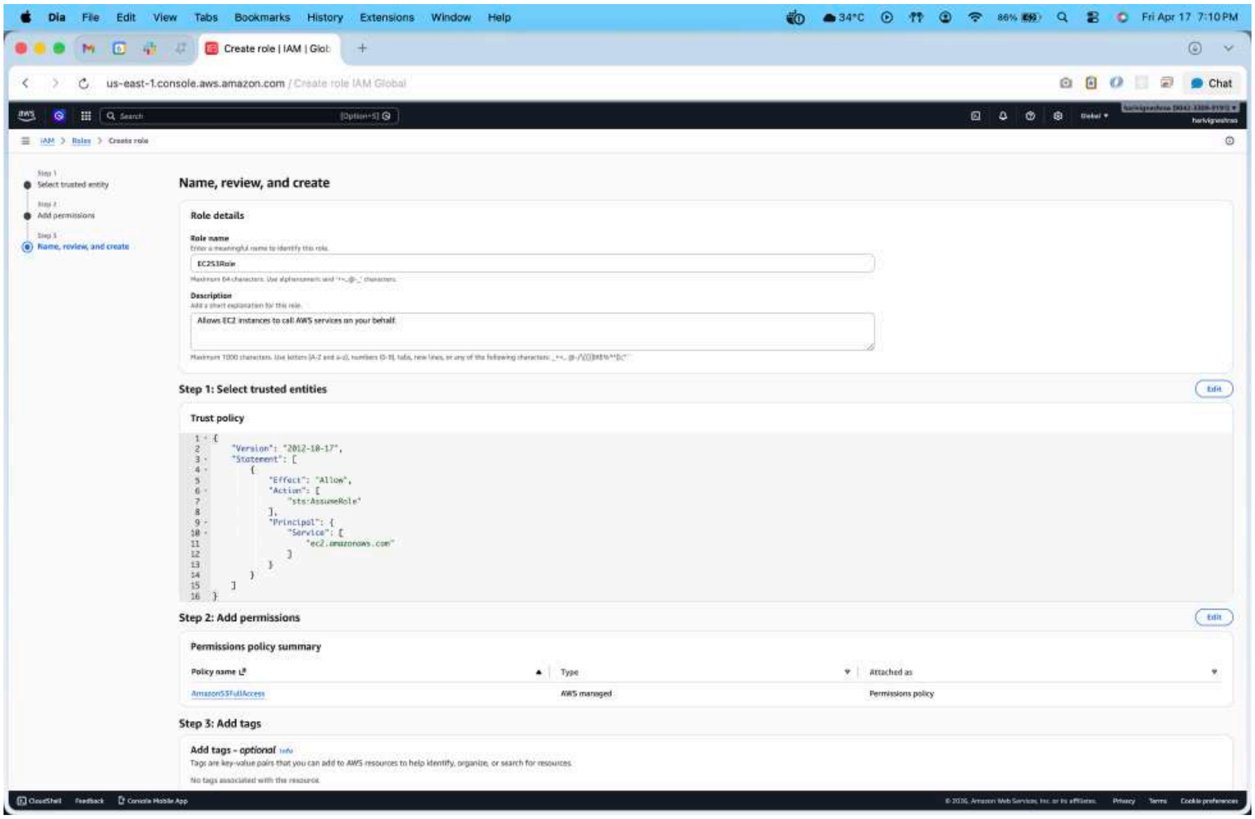
Step 2 — Add Permissions

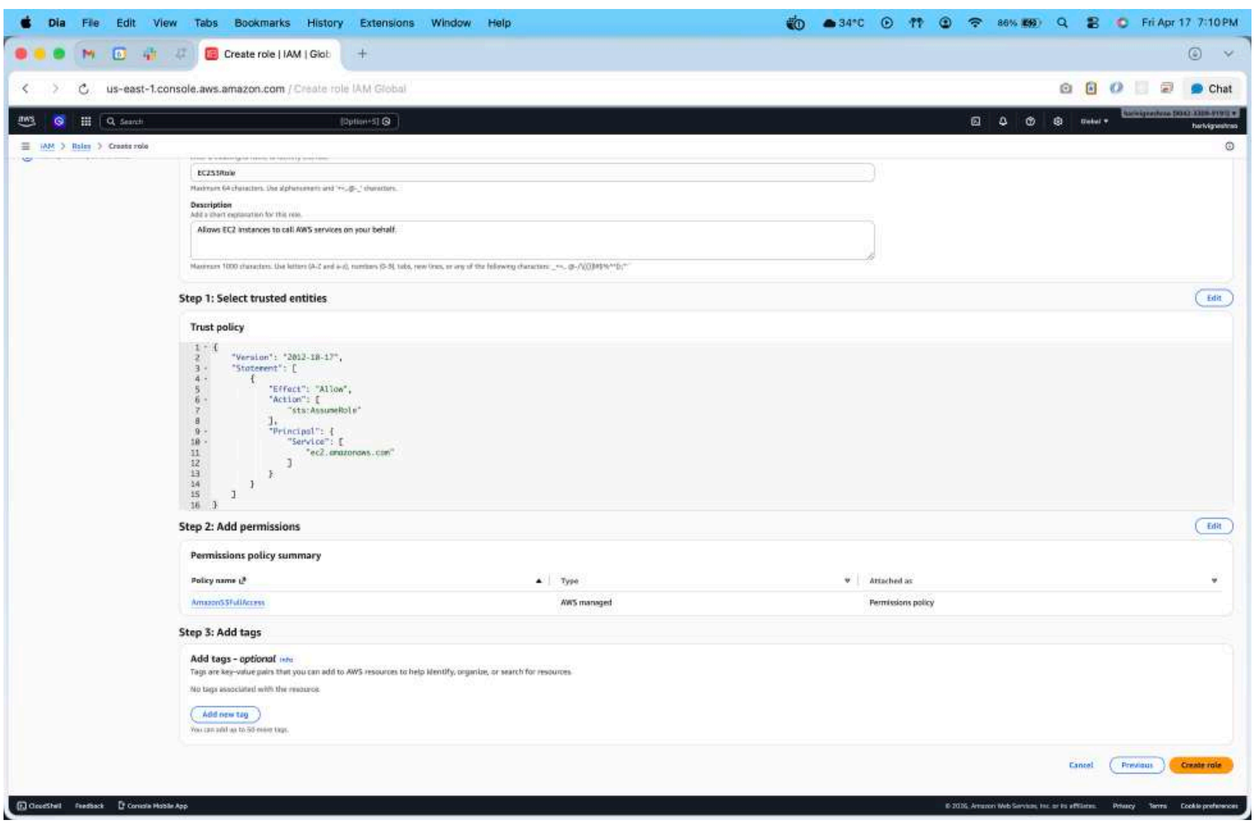
1. Search policy: AmazonS3FullAccess
2. Select it → Next



Step 3 — Name the Role

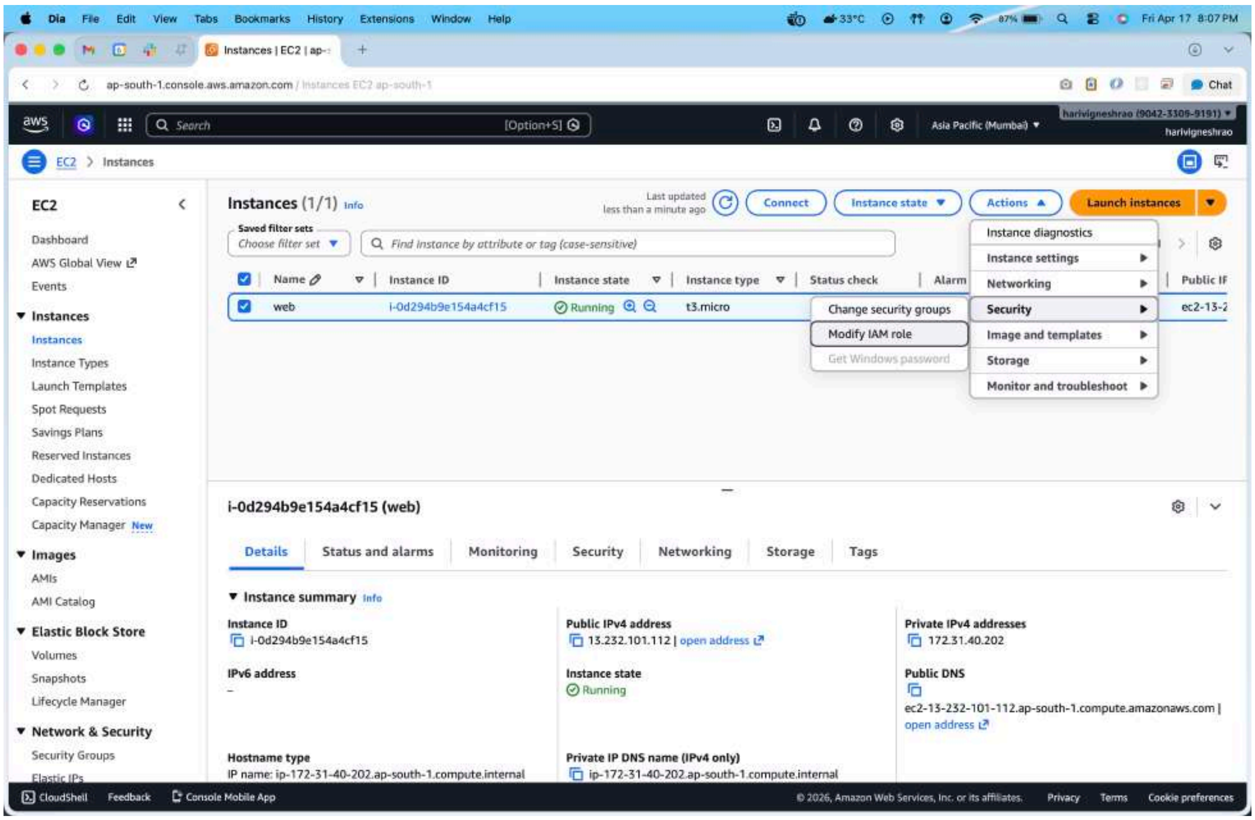
- Name: EC2-S3-Role
- Click Create role

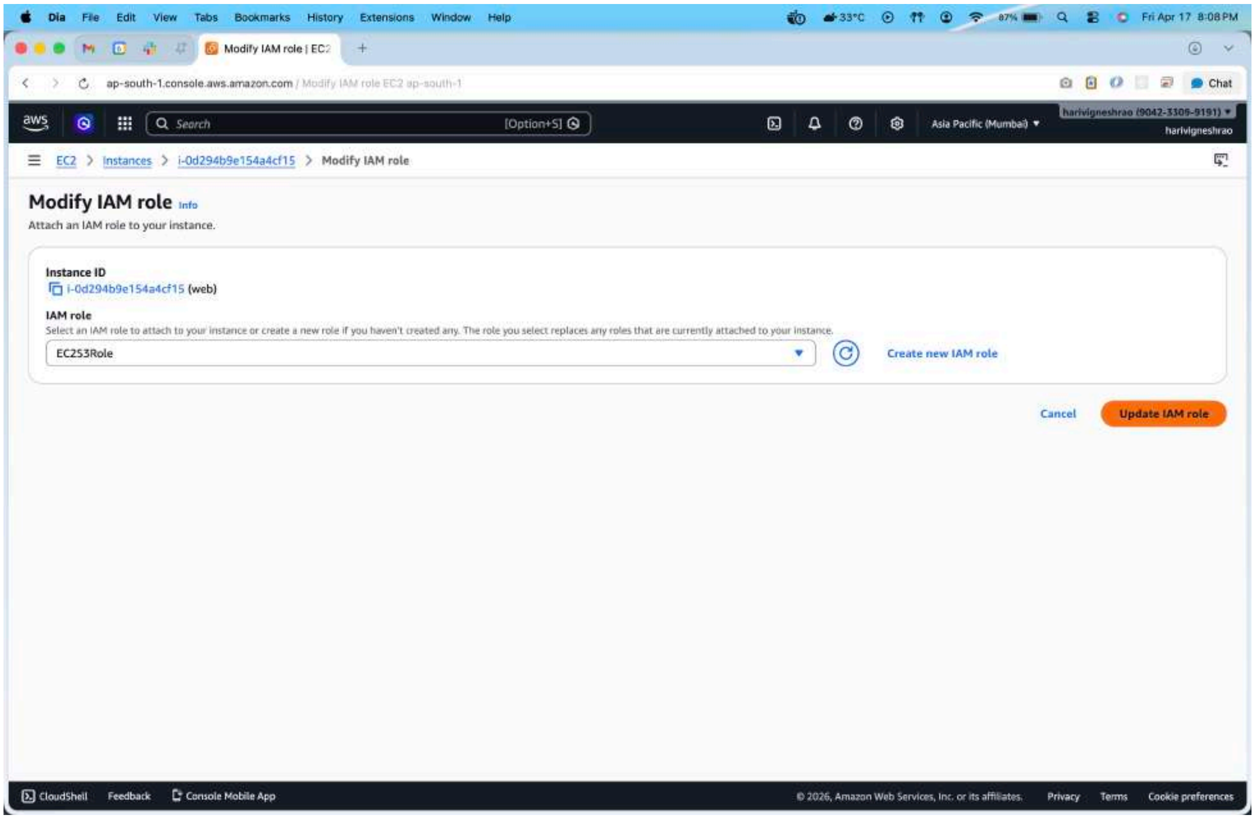




Step 4 — Attach Role to EC2 Instance

1. Go to EC2 Dashboard
2. Select running instance
3. Actions → Security → Modify IAM role
4. Select EC2-S3-Role → Update





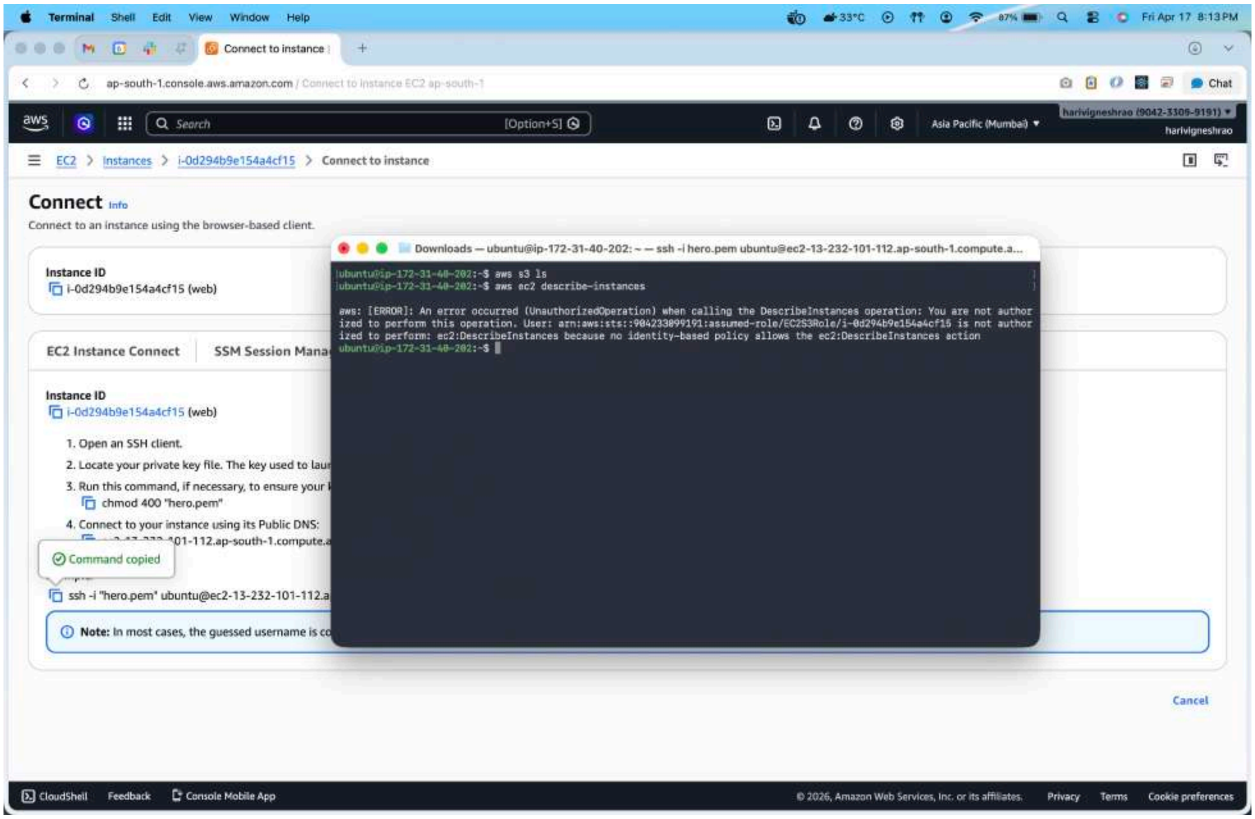
Step 5 — Verify from Instance (No Access Keys!)

SSH into instance and run:

aws s3 ls
aws ec2 describe-instances

- aws s3 ls Works
- aws ec2 describe-instances **Unauthorized (no EC2 permission)**

Reason: Instance receives **temporary credentials** from IAM Role automatically.


Terminal Shell Edit View Window Help
Fri Apr 17 8:13 PM
Connect to instance
ap-south-1.console.aws.amazon.com
Chat
Search [Option+S]
Asia Pacific (Mumbai)
harivigneshrao
EC2 > Instances > i-0d294b9e154a4cf15 > Connect to instance
Connect Info
Connect to an instance using the browser-based client.
Instance ID
i-0d294b9e154a4cf15 (web)
EC2 Instance Connect
SSM Session Mana
Instance ID
i-0d294b9e154a4cf15 (web)
1. Open an SSH client.
2. Locate your private key file. The key used to lau
3. Run this command, if necessary, to ensure your
chmod 400 "hero.pem"
4. Connect to your instance using its Public DNS:
01-112.ap-south-1.compute.a
Command copied
ssh -i "hero.pem" ubuntu@ec2-13-232-101-112.a
Note: In most cases, the guessed username is co
Downloads — ubuntu@ip-172-31-40-202: ~ — ssh -i hero.pem ubuntu@ec2-13-232-101-112.ap-south-1.compute.a...
ubuntu@ip-172-31-40-202:~$ aws s3 ls
ubuntu@ip-172-31-40-202:~$ aws ec2 describe-instances
aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not author
ized to perform this operation. User: arn:aws:sts::904233099191:assumed-role/EC2S3Role/i-0d294b9e154a4cf15 is not author
ized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
ubuntu@ip-172-31-40-202:~$
Cancel
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